W10-HW

Instructions

- Do not delete, edit, or change any text or questions in this document.
- Type or write your answers directly below each question.
- Show all work. If you submit images, make sure they are correctly oriented (not sideways or upside down) and easy to read.
- **Submit your work only through Blackboard. Do not email it**.
- If you submit photos of handwritten work, ensure your writing is clear and legible, and that the images are large enough to read easily.
- Answer all questions in complete sentences and include enough detail to clearly show your understanding and reasoning.
- Assignments that do not show all required work will receive a grade of 0
- Clearly explain your reasoning at each step.
- Demonstrate each step of RSA key generation, encryption, and decryption.
  Not following these instruction will result in a grade penalty.

Part 1: Choose Your Prime Numbers 10 Points

1. Choose two small prime numbers p and q. Keep them small so calculations are manageable.
   Examples: 3, 5, 7, 11, 13, 17

   Your choice:
   p = _____
   q = _____

Part 1

1 $p = 7$ $q = 11$

2. Compute the RSA modulus:
   n = p × q = _____

2 $n = 7 \cdot 11 = 77$

3. Compute Euler's totient value:
$\varphi(n) = (p - 1)(q - 1) =$ _____

③ $\varphi(n) = (7-1)\cdot(11-1) = 6 \cdot 10 = 60$

Part 2: Choose the Public Exponent 15 Points

4. Choose a value for the public exponent e such that:
$\gcd(e, \varphi(n)) = 1$

Your choice:
e = _____

④ $e = 7$

5. Verify that $\gcd(e, \varphi(n)) = 1$ by showing your calculation.

⑤ $\gcd(7, 60)$
$60/7 = 8 r 4$
$7/4 = 1 r 3$
$4/3 = 1 r 1$
$3/1 = 3 r 0$

$\gcd = 1$

Part 3: Compute the Private Key 15 Points

6. Compute the private exponent d such that:
$e \times d \equiv 1 \pmod{\varphi(n)}$

Show how you found the modular inverse.

d = _____

$4/3 = 1 \; r1$

$3/1 = 3 \; r0$

$\gcd = 1$

$d = 43$

$d = -17 \bmod 60$

$-17 \cdot 7 = 1 \bmod 60$

$1 = 2(60) + (-17)(7)$

Part 3

6) $7d = 1 \pmod{60}$ $\quad 1 = 2(60) - 17(7)$

$7d + 60k = 1$ $\quad 1 = 2(60) - 16(7) - 7$

$60 = 8(7) + 4 \rightarrow 1 = 2(60 - 8(7))$

$7 = 1(4) + 3 \rightarrow 1 = 4 - (7 - 1(4))$; $1 = 2(4) - 7$

$4 = 1(3) + 1 \rightarrow 1 = 4 - 1(3)$

$3 = 3(1) + 0$

7. Write your keys:

Public Key: (n, e) = _____
Private Key: (n, d) = _____

7) Pub key$(n, e) = (77, 7)$ Priv key$(n, d) = (77, 43)$

Part 4: Encryption **20 Points**

8. Choose a small message m such that m < n.

m = _____

Part 4

(8) [redacted] $m = 9$

9. Encrypt the message using your public key:

c = m^e mod n

Show your calculation and compute the ciphertext.

(9) $c = m^{\wedge e} \bmod n$

$c = 9^7 \bmod 77$ $\quad$ $c = 576 \bmod 77$

$9^{\wedge}2 = 81 \equiv 4 \% 77$ $\quad$ $c = 37$

$9^{\wedge}4 \equiv 4^{\wedge}2 = 16 \% 77$

$9^{\wedge}7 = 9 \cdot 9^{\wedge}2 \cdot 9^{\wedge}4$

$= 9 \cdot 4 \cdot 16$

$= 576$

Part 5: Decryption 20 Points

10. Decrypt the ciphertext using your private key:

m = c^d mod n

Show your work and verify that you recover the original message.

Part 5

10 $m = c^d \bmod n$

$m = 37^{43} \bmod 77$

$37^2 = 1369 = 60 \% 77$

$37^4 = 60^2 = 3600 = 58 \% 77$

$37^8 = 58^2 = 3364 = 53 \% 77$

$37^{16} = 53^2 = 2809 = 37 \% 77$

$37^{32} = 37^2 = 60 \% 77$

$= 60 \cdot 53 \cdot 60 \cdot 37$

$60 \cdot 53 = 3180 = 23 \% 77$

$23 \cdot 60 = 1380 = 71 \% 77$

$71 \cdot 37 = 2627 = 9 \% 77$

$m = 9$

Part 6: Conceptual Questions 20 Points

11. Why must p and q be prime numbers in RSA?

q & p must be prime numbers because the mathematical properties of RSA may not work properly with non-prime numbers which is especially important when calculating Euler's totient function.

12. Why must gcd(e, φ(n)) = 1?

e must be relatively prime to φ(n) so that the mod inverse of d can exist as without it the private key can't be computed and decryption wouldn't be possible.

13. Why are very large primes used in real RSA systems instead of the small primes used in this exercise?

Large primes are used because they make it much more difficult to factor n into p & q which ensures stronger security as factoring the large numbers is very expensive.

14. Explain why RSA encryption is slower than symmetric encryption algorithms such as AES.

RSA is slower because of the complex mathematical operations such as mod exponentation with the large numbers. AES however usees simpler operations that are faster and more efficient when processing large amounts of data.